CITIBANK APPLIED RESEARCH PROJECT · GROUP 1

Evaluating Human-in-the-Loop LLM Workflows for Market-Relevant Earnings Analysis

Bayes Business School

MSc Business Analytics

Deadline: 31 Aug 2025

SECTION 1

Context & Background

Who is behind this project, who it's for, and the problem it's trying to solve.

Industry Partner	Citibank — multinational financial services institution (investment banking, retail banking, data-driven solutions)
Supervisors	Dr Jan Rock (SVP, Citibank) · Dr Ahmad Abu-Khazneh (City University)
Core Problem	Financial markets depend on unstructured text — earnings reports, Fed minutes, analyst calls. LLMs offer the ability to process these at scale, but reliability, evaluation, and workflow integration remain open and unresolved questions.
Target End Users	Financial analysts, traders, institutional decision-makers, FinTech startups
Student Skill Set	Python · Machine learning · NLP / LLMs · Financial markets & unstructured data

SECTION 2

Research Question & Project Goal

The central question we're answering and the concrete goal we're working towards.

> **RESEARCH QUESTION**
>
> Can a human-in-the-loop LLM workflow improve the speed, consistency, and reliability of extracting market-relevant sentiment, risk signals, and forward guidance from corporate earnings documents — compared with human-only, LLM-only, and traditional NLP approaches?

> **PROJECT GOAL**
>
> Build and evaluate a prototype system that transforms earnings documents into structured, evidence-backed analyst briefings, then test whether it improves financial-text analysis under time pressure.

> **THE ARGUMENT WE'RE MAKING**
>
> LLMs should not be treated as autonomous trading systems. However, they can materially improve the speed and structure of financial document analysis by producing evidence-backed first-pass analyst briefings. The optimal workflow is likely hybrid: the LLM extracts and organises market-relevant signals, while a human validates interpretation, resolves ambiguity, and makes the final judgement.

SECTION 3

Deliverable 1 — Dataset

The raw material the project runs on. A curated set of real earnings documents with metadata that allows us to link LLM outputs back to actual market events.

Target Size	30–40 documents across 6–8 companies, 4 quarters each
Minimum	20 documents across 5 companies, 4 quarters each
Event Mix	Positive, negative, and neutral earnings events across different sectors
Metadata	Company · Quarter · Release date/time · Document type · Sector · Source URL · Next-day return (where available) · Notes on major market reaction

SECTION 4

Deliverable 2 — Structured LLM Extraction Pipeline

The core technical component. Takes a raw earnings document as input and produces a structured analyst briefing. Every output claim must link back to a quote from the source — no unsupported assertions.

Input: earnings release or call transcript. Output fields:

• Executive Summary	• Overall Sentiment
• Sentiment Confidence	• Key Positive Signals
• Key Negative Signals	• Forward Guidance
• Risk Categories	• Market-Moving Statements
• Evidence Quotes (source-linked)	• Human-Review Flags

SECTION 5

Deliverable 3 — Human Audit Workflow

What makes this a human-in-the-loop system rather than a fully automated one. A reviewer checks each LLM claim and applies one of four labels — this is what we evaluate, and what gives the system credibility in a financial context.

Label	Meaning
Accept	LLM claim is correct and well-supported by the source document
Edit	Partially correct — the claim needs revision before it can be used
Reject	Claim is unsupported or factually wrong — discard
Unclear	Cannot determine accuracy without further verification

Format: this does not need to be complex — a spreadsheet, Jupyter notebook, or simple Streamlit app will work.

SECTION 6

Deliverable 4 — Evaluation Framework

How we measure whether our system is actually better than the alternatives. Each metric targets a different dimension of performance, covering quality, cost, speed, and practical usefulness.

Metric	How We Measure It
Accuracy	Human / expert rubric score against ground truth
Completeness	% of key points from the document that are captured
Evidence Support	% of LLM claims backed by a direct source quote
Hallucination Rate	% of claims that are unsupported or fabricated

Speed	Minutes taken to process one document end-to-end
Consistency	Agreement across repeated runs or different prompt formulations
Cost	API cost (£ or $) per document processed
Usefulness	Qualitative rating from the human reviewer (1–5 scale)

SECTION 7

Deliverable 5 — Baseline Comparison

To make meaningful claims about our system, we need to compare it against alternatives. These four methods form the comparison set — we run the same evaluation framework across all of them.

Method	Role in This Project
Human-Only	Current manual process. Sets the quality benchmark — what a skilled analyst produces without any AI assistance.
Traditional NLP / FinBERT	Established NLP baseline. Shows what rule-based and fine-tuned models can do without LLMs.
LLM-Only (fully automated)	No human review step. Tests the ceiling of raw LLM performance and exposes where hallucination is a risk.
Hybrid LLM + Human (ours)	Our proposed system. LLM handles extraction and structuring; human validates and makes the final call.

SECTION 8

Deliverable 6 — Optional Market-Reaction Case Study

An exploratory extension if time and data allow. Not a core requirement — the project does not stand or fall on this.

For a subset of documents, compare LLM-extracted signals against real market outcomes: next-day return, volatility, abnormal return, and analyst / news commentary. Treat as exploratory evidence only — weak predictive performance does not undermine the project.

SECTION 9

Key Dates & Deliverable Schedule

The project timeline as set out in the brief.

Apr 2025	Familiarise with project brief, industry partner, and relevant literature
1 May 2025	Submit realistic project proposal
May–Aug 2025	Project execution — build system, collect data, run evaluations
31 Aug 2025	Academic report (3,500–5,000 words) submitted. Business summary delivered to Citibank.